

```

(kali㉿kali)-[~/volatility3]
$ ./vol.py -h
Volatility 3 Framework 2.28.2
usage: vol.py [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]] [-e EXTEND] [-p PLUGIN_DIRS]
              [-s SYMBOL_DIRS] [-v] [-l LOG] [-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
              [--save-config SAVE_CONFIG] [--clear-cache] [--cache-path CACHE_PATH] [--offline] [-u URL]
              [--filters FILTERS] [--hide-columns [HIDE_COLUMNS ...]] [-r RENDERER]
              [--single-location SINGLE_LOCATION] [--stackers [STACKERS ...]]
              [--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]]
              PLUGIN ...

An open-source memory forensics framework

options:
  -h, --help                Show this help message and exit, for specific plugin options use 'vol.py <pluginname>
                           --help'
  -c, --config CONFIG        Load the configuration from a json file
  --parallelism [{processes,threads,off}]
                           Enables parallelism (defaults to off if no argument given)
  -e, --extend EXTEND        Extend the configuration with a new (or changed) setting
  -p, --plugin-dirs PLUGIN_DIRS
                           Semi-colon separated list of paths to find plugins
  -s, --symbol-dirs SYMBOL_DIRS
                           Semi-colon separated list of paths to find symbols
  -v, --verbosity            Increase output verbosity

```

```

-l, --log LOG                Log output to a file as well as the console
-o, --output-dir OUTPUT_DIR
                           Directory in which to output any generated files
-q, --quiet                 Remove progress feedback
-f, --file FILE              Shorthand for --single-location=file:// if single-location is not defined
--write-config               Write configuration JSON file out to config.json
--save-config SAVE_CONFIG    Save configuration JSON file to a file
--clear-cache                Clears out all short-term cached items
--cache-path CACHE_PATH      Change the default path (/home/kali/.cache/volatility3) used to store the cache
--offline                    Do not search online for additional JSON files
-u, --remote-isf-url URL      Search online for ISF json files
--filters FILTERS            List of filters to apply to the output (in the form of [+<columnname,pattern[!]])
--hide-columns [HIDE_COLUMNS ...]
                           Case-insensitive space separated list of prefixes to determine which columns to hide
                           in the output if provided
-r, --renderer RENDERER      Determines how to render the output (quick, none, csv, pretty, json, jsonl, mermaid)
--single-location SINGLE_LOCATION
                           Specifies a base location on which to stack
--stackers [STACKERS ...]    List of stackers
--single-swap-locations [SINGLE_SWAP_LOCATIONS ...]

```

```

(kali㉿kali)-[~/volatility3]
$ git clone https://github.com/volatilityfoundation/volatility.git
ls volatility
cd volatility
Cloning into 'volatility' ...
remote: Enumerating objects: 27414, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 27414 (delta 0), reused 0 (delta 0), pack-reused 27412 (from 2)
Receiving objects: 100% (27414/27414), 21.11 MiB | 2.17 MiB/s, done.
Resolving deltas: 100% (19758/19758), done.
AUTHORS.txt  CREDITS.txt  Makefile      pyinstaller  resources  volatility
CHANGELOG.txt LEGAL.txt     MANIFEST.in  pyinstaller.spec  setup.py   vol.py
contrib      LICENSE.txt  PKG-INFO     README.txt     tools
(kali㉿kali)-[~/volatility3/volatility]
$ python2 vol.py --info
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)

```

```

** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.evttlogs (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.ssdtd (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
** Failed to import volatility.plugins.envvars (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)

```

#### Plugins

```

atoms          - Print session and window station atom tables
atomscan       - Pool scanner for atom tables
bigpools       - Dump the big page pools using BigPagePoolScanner
bioskbd        - Reads the keyboard buffer from Real Mode memory
callbacks      - Print system-wide notification routines
clipboard      - Extract the contents of the windows clipboard

```

```

LinuxAMD64PagedMemory - Linux-specific AMD 64-bit address space.
MachOAddressSpace    - Address space for mach-o files to support atc-ny memory reader
OSXPmemELF           - This AS supports VirtualBox ELF64 coredump format
QemuCoreDumpElf      - This AS supports Qemu ELF32 and ELF64 coredump format
SkipDuplicatesAMD64PagedMemory - Windows 8/10-specific AMD 64-bit address space.
VMWareAddressSpace   - This AS supports VMware snapshot (VMSS) and saved state (VMSS) files
VMWareMetaAddressSpace - This AS supports the VMEM format with VMSN/VMSS metadata
VirtualBoxCoreDumpElf64 - This AS supports VirtualBox ELF64 coredump format
WindowsAMD64PagedMemory - Windows-specific AMD 64-bit address space.
WindowsCrashDumpSpace32 - This AS supports windows Crash Dump format
WindowsCrashDumpSpace64 - This AS supports windows Crash Dump format
WindowsCrashDumpSpace64BitMap - This AS supports Windows BitMap Crash Dump format
WindowsHiberFileSpace32 - This is a hibernate address space for windows hibernation files.

```

#### Scanner Checks

```

CheckPoolSize      - Check pool block size
CheckPoolType      - Check the pool type
KPCRScannerCheck   - Checks the self referential pointers to find KPCRs
MultiPrefixFinderCheck - Checks for multiple strings per page, finishing at the offset
MultiStringFinderCheck - Checks for multiple strings per page
PoolTagCheck       - This scanner checks for the occurrence of a pool tag

```

```

(kali@kali)-[~/volatility3/volatility]

```

```

(kali@kali)-[~/volatility3/volatility]
$ wget https://www.dropbox.com/s/x41f2lyhlrixts6/memdumpWin7.mem
--2026-09-14 11:51:30-- https://www.dropbox.com/s/x41f2lyhlrixts6/memdumpWin7.mem
Resolving www.dropbox.com (www.dropbox.com)... 162.125.64.18, 2620:100:6024:18::a27d:4412
Connecting to www.dropbox.com (www.dropbox.com)|162.125.64.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/sc/fi/1y68wpuqi1fsvrapq9egg/memdumpWin7.mem?rlkey=pk3thrk3p2fk9m90mywksx3s1 [
following]
--2026-09-14 11:51:31-- https://www.dropbox.com/sc/fi/1y68wpuqi1fsvrapq9egg/memdumpWin7.mem?rlkey=pk3thrk3p2fk
9m90mywksx3s1
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://uca793e97106dfd7e41aeec42499.dl.dropboxusercontent.com/cd/0/inline/DIE8EiflGVUgb3uW0iIypYK5T0F
-lqWF4jftQcMCKkDY9s-u03C0QtClK7UThyzHyHIFR5Hv-bYH4u68arx7PzMQzIuzNip0kz6wTt8jt3Egz8_ojt5gKqDrAYBkHosjuR0/file# [
following]
--2026-09-14 11:51:32-- https://uca793e97106dfd7e41aeec42499.dl.dropboxusercontent.com/cd/0/inline/DIE8EiflGVUg
b3uW0iIypYK5T0F-lqWF4jftQcMCKkDY9s-u03C0QtClK7UThyzHyHIFR5Hv-bYH4u68arx7PzMQzIuzNip0kz6wTt8jt3Egz8_ojt5gKqDrAYBk
HosjuR0/file
Resolving uca793e97106dfd7e41aeec42499.dl.dropboxusercontent.com (uca793e97106dfd7e41aeec42499.dl.dropboxusercon
tent.com)... 162.125.64.15, 2620:100:6024:15::a27d:440f
Connecting to uca793e97106dfd7e41aeec42499.dl.dropboxusercontent.com (uca793e97106dfd7e41aeec42499.dl.dropboxuse
rcontent.com)|162.125.64.15|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DIE91tb9nu0gCQceI20lwV6lAkGKG7YEFohXMrXCA0o4zeBRP4oIa0FxCp6CNfV2vpmHxCNCv8Xu7yJ820yhXAJN
smTxLjUqB8xtyCvM5PRIHAUoJrSWcwwk-h0azj3wnLDPRsOfntYdQDdoEBQp16mL0qxhg-a5A3ZjCqGr5bTddB8_naPl97sc5dgIpyccEMT03GT
TJ6DSNhKvz5xqY00Vgaz3X0q0D SrTEZSeEi4S0smp1prHCZULcfFEkI979LLNIRgddZYeI8xJsPCqxI1CfRvN-v0SjdiD8 n04d2nUFj0D9Hwd

```

```
tent.com)... 162.125.64.15, 2620:100:6024:15::a27d:440f
Connecting to uca03782f8d8dcc01c1ace857cb7.dl.dropboxusercontent.com (uca03782f8d8dcc01c1ace857cb7.dl.dropboxuse
rcontent.com)|162.125.64.15|:443 ... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DINTuUf8KruixuEa24-iYnlb2vhc7kZkfcbF6UX60k1LoNugNFScf4RZlSyAdGGVUrmk0HFHXH1Nrr-aK3IVK1JEv
3_DADadTAacBmMv7XXwLmm4jwrDUHQx5Re_umKSapv_UWL7o6f8d3fiz_-tnzNCXdm9TQrTL7_1Zf2nveKI74sNk4MsEC41rth1HVMr1BJRsShLE
m3oIVFNLjdTXAO_j69P_t3a9rHKIKdYkHR-mLLtkdo3z7W7G7zaHl1pyvWt-Gil6vDg5Tt6yqvYrMJB1VDnnHDzYxfgTNOUFPME8cl_IE6Y_Jb
SAmtHD-ZZZmAtbXVX3n2k_z-Ptgi7wijXAuvXduCeNJApt_4cFyBA/file [following]
--2026-09-16 05:00:00-- https://uca03782f8d8dcc01c1ace857cb7.dl.dropboxusercontent.com/cd/0/inline2/DINTuUf8Kru
ixuEa24-iYnlb2vhc7kZkfcbF6UX60k1LoNugNFScf4RZlSyAdGGVUrmk0HFHXH1Nrr-aK3IVK1JEv3_DADadTAacBmMv7XXwLmm4jwrDUHQx5Re_
umKSapv_UWL7o6f8d3fiz_-tnzNCXdm9TQrTL7_1Zf2nveKI74sNk4MsEC41rth1HVMr1BJRsShLEm3oIVFNLjdTXAO_j69P_t3a9rHKIKdYkHR
-mLLtkdo3z7W7G7zaHl1pyvWt-Gil6vDg5Tt6yqvYrMJB1VDnnHDzYxfgTNOUFPME8cl_IE6Y_JbSAmtHD-ZZZmAtbXVX3n2k_z-Ptgi7wijXAu
vXduCeNJApt_4cFyBA/file
Reusing existing connection to uca03782f8d8dcc01c1ace857cb7.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1073676288 (1024M) [application/octet-stream]
Saving to: 'memdumpWin7.mem'
```

```
memdumpWin7.mem          100%[=====>] 1024M 2.62MB/s in 9m 1s

2026-09-16 05:10:04 (1.89 MB/s) - 'memdumpWin7.mem' saved [1073676288/1073676288]
```

```
(root@kali)-[/home/kali/volatility3/volatility]
```

```
(root@kali)-[/home/kali/volatility3/volatility]
# ls -l
total 1048516
-rw-r--r-- 1 root root 1073676288 Sep 16 05:10 memdumpWin7.mem
```

```
(root@kali)-[/home/kali/volatility3]
# python3 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem windows.info
Volatility 3 Framework 2.28.2
Progress: 0.12 Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 0.2439C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 0.3539C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 0.4739C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 0.5939C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 0.7139C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 0.8239C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 0.9439C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 1.0639C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 1.1839C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 1.3039C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 1.4139C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 1.5339C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 1.6539C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 1.7739C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 1.8839C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 2.0039C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 2.1239C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 2.2439C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 2.3639C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 2.4739C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 2.5939C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
Progress: 2.7139C22/ntkrpamp.Reading file http://msdl.microsoft.com/download/symbols/ntkrpamp.pdb/2F6693D273A
```

| Variable                       | Value                                                                                                        |
|--------------------------------|--------------------------------------------------------------------------------------------------------------|
| Kernel Base                    | 0x82847000                                                                                                   |
| DTB                            | 0x185000                                                                                                     |
| Symbols file                   | ///home/kali/volatility3/volatility3/symbols/windows/ntkrpamp.pdb/2F6693D273A446AA8740983BC3D539C2-2.json.xz |
| Is64Bit                        | False                                                                                                        |
| IsPAE                          | True                                                                                                         |
| layer_name                     | 0 WindowsIntelPAE                                                                                            |
| memory_layer                   | 1 FileLayer                                                                                                  |
| KdDebuggerDataBlock            | 0x82972c68                                                                                                   |
| NTBuildLab                     | 7601.18939.x86fre.win7sp1_gdr.15                                                                             |
| CSDVersion                     | 1                                                                                                            |
| KdVersionBlock                 | 0x82972c40                                                                                                   |
| Major/Minor                    | 15.7601                                                                                                      |
| MachineType                    | 332                                                                                                          |
| KeNumberProcessors             | 1                                                                                                            |
| SystemTime                     | 2019-01-06 15:09:06+00:00                                                                                    |
| NtSystemRoot                   | C:\Windows                                                                                                   |
| NtProductType                  | NtProductWinNt                                                                                               |
| NtMajorVersion                 | 6                                                                                                            |
| NtMinorVersion                 | 1                                                                                                            |
| PE MajorOperatingSystemVersion | 6                                                                                                            |

```
PE MajorOperatingSystemVersion 6
PE MinorOperatingSystemVersion 1
PE Machine 332
PE TimeDateStamp Wed Jul 22 16:46:25 2015
```

```
(root@kali)-[/home/kali/volatility3]
#
```

```
INFO : volatility.debug : Determining profile based on KDBG search...
      Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
      AS Layer1 : IA32PagedMemoryPae (Kernel AS)
      AS Layer2 : FileAddressSpace (/home/kali/volatility3/volatility/memdumpWin7.mem)
      PAE type : PAE
      DTB : 0x185000L
      KDBG : 0x82972c68L
      Number of Processors : 1
      Image Type (Service Pack) : 1
      KPCR for CPU 0 : 0x82973d00L
      KUSER_SHARED_DATA : 0xffdf0000L
      Image date and time : 2019-01-06 15:09:06 UTC+0000
      Image local date and time : 2019-01-06 07:09:06 -0800
```

```
(root@kali)-[/home/kali/volatility]
#
```

Legend: (S) = Stable (V) = Volatile

```
Registry: \REGISTRY\MACHINE\BCD00000000
Key name: NewStoreRoot (S)
Last updated: 2019-01-06 18:02:44 UTC+0000
```

Subkeys:

- (S) Description
- (S) Objects

Values:

```
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: CMI-CreateHive{6A1C4018-979D-4291-A7DC-7AED1C75B67C} (S)
Last updated: 2019-01-06 15:02:48 UTC+0000
```

Subkeys:

- (S) AppEvents
- (S) Console
- (S) Control Panel
- (S) Environment
- (S) EUDC
- (S) Keyboard Layout
- (S) Network
- (S) Printers

Legend: (S) = Stable (V) = Volatile

Registry: \??\C:\Users\IEUser\ntuser.dat

Key name: Volatile Environment (V)

Last updated: 2019-01-06 15:02:48 UTC+0000

Subkeys:

(V) 1

Values:

|        |              |   |     |                                 |
|--------|--------------|---|-----|---------------------------------|
| REG_SZ | LOGONSERVER  | : | (V) | \\IE8WIN7                       |
| REG_SZ | USERDOMAIN   | : | (V) | IE8WIN7                         |
| REG_SZ | USERNAME     | : | (V) | IEUser                          |
| REG_SZ | USERPROFILE  | : | (V) | C:\Users\IEUser                 |
| REG_SZ | HOMEPATH     | : | (V) | \Users\IEUser                   |
| REG_SZ | HOMEDRIVE    | : | (V) | C:                              |
| REG_SZ | APPDATA      | : | (V) | C:\Users\IEUser\AppData\Roaming |
| REG_SZ | LOCALAPPDATA | : | (V) | C:\Users\IEUser\AppData\Local   |

(root@kali)-[/home/kali/volatility]  
#

Legend: (S) = Stable (V) = Volatile

Registry: \SystemRoot\System32\Config\SOFTWARE

Key name: ProfileList (S)

Last updated: 2015-09-21 09:50:52 UTC+0000

Subkeys:

(S) S-1-5-18  
(S) S-1-5-19  
(S) S-1-5-20  
(S) S-1-5-21-1716914095-909560446-1177810406-1000  
(S) S-1-5-21-1716914095-909560446-1177810406-1002

Values:

|               |                   |   |     |                             |
|---------------|-------------------|---|-----|-----------------------------|
| REG_EXPAND_SZ | ProfilesDirectory | : | (S) | %SystemDrive%\Users         |
| REG_EXPAND_SZ | Default           | : | (S) | %SystemDrive%\Users\Default |
| REG_EXPAND_SZ | Public            | : | (S) | %SystemDrive%\Users\Public  |
| REG_EXPAND_SZ | ProgramData       | : | (S) | %SystemDrive%\ProgramData   |

(root@kali)-[/home/kali/volatility]  
#

Legend: (S) = Stable (V) = Volatile

Registry: \SystemRoot\System32\Config\SOFTWARE

Key name: S-1-5-21-1716914095-909560446-1177810406-1002 (S)

Last updated: 2015-09-21 09:51:49 UTC+0000

Subkeys:

Values:

REG\_EXPAND\_SZ ProfileImagePath : (S) C:\Users\sshd\_server

REG\_DWORD Flags : (S) 0

REG\_DWORD State : (S) 256

REG\_BINARY Sid : (S)

0x00000000 01 05 00 00 00 00 00 05 15 00 00 00 af 07 56 66 .....Vf

0x00000010 7e ca 36 36 e6 f5 33 46 ea 03 00 00 ~.66..3F....

REG\_DWORD ProfileLoadTimeLow : (S) 0

REG\_DWORD ProfileLoadTimeHigh : (S) 0

REG\_DWORD RefCount : (S) 1

(root@kali)-[/home/kali/volatility]

Legend: (S) = Stable (V) = Volatile

Registry: \SystemRoot\System32\Config\SOFTWARE

Key name: S-1-5-21-1716914095-909560446-1177810406-1000 (S)

Last updated: 2019-01-06 15:07:15 UTC+0000

Subkeys:

Values:

REG\_EXPAND\_SZ ProfileImagePath : (S) C:\Users\IEUser

REG\_DWORD Flags : (S) 0

REG\_DWORD State : (S) 0

REG\_BINARY Sid : (S)

0x00000000 01 05 00 00 00 00 00 05 15 00 00 00 af 07 56 66 .....Vf

0x00000010 7e ca 36 36 e6 f5 33 46 e8 03 00 00 ~.66..3F....

REG\_DWORD ProfileLoadTimeLow : (S) 0

REG\_DWORD ProfileLoadTimeHigh : (S) 0

REG\_DWORD RefCount : (S) 3

REG\_DWORD RunLogonScriptSync : (S) 0

(root@kali)-[/home/kali/volatility]

```
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: Winlogon (S)
Last updated: 2019-01-06 15:02:47 UTC+0000
```

```
Subkeys:
(S) GPEExtensions
(V) AutoLogonChecked
```

```
Values:
REG_SZ      ReportBootOk      : (S) 1
REG_SZ      Shell              : (S) explorer.exe
REG_SZ      PreCreateKnownFolders : (S) {A520A1A4-1780-4FF6-BD18-167343C5AF16}
REG_SZ      Userinit           : (S) C:\Windows\system32\userinit.exe,
REG_SZ      VMApplet           : (S) SystemPropertiesPerformance.exe /pagefile
REG_SZ      WinStationsDisabled : (S) 0
REG_DWORD   DisableCAD        : (S) 1
REG_SZ      scremoveoption     : (S) 0
REG_DWORD   ShutdownFlags     : (S) 2147483687
REG_SZ      DefaultDomainName  : (S)
REG_SZ      DefaultUserName    : (S) IEUser
REG_SZ      AutoAdminLogon     : (S) 1
```

```
(root@kali)-[/home/kali/volatility]
#
```

| Virtual    | Physical   | Name                                                                 |
|------------|------------|----------------------------------------------------------------------|
| 0x87c10370 | 0x27fd5370 | [no name]                                                            |
| 0x87c1c008 | 0x27fa3008 | \REGISTRY\MACHINE\SYSTEM                                             |
| 0x87c459c8 | 0x27d8e9c8 | \REGISTRY\MACHINE\HARDWARE                                           |
| 0x889c0430 | 0x1285a430 | ??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT              |
| 0x8b46d008 | 0x16d8c008 | ??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT                |
| 0x8b5d39b0 | 0x120049b0 | ??\C:\Users\IEUser\ntuser.dat                                        |
| 0x97104008 | 0x1c45f008 | \SystemRoot\System32\Config\SECURITY                                 |
| 0x9711d9c8 | 0x1e6a09c8 | \SystemRoot\System32\Config\SOFTWARE                                 |
| 0x981f9008 | 0x1b446008 | \SystemRoot\System32\Config\DEFAULT                                  |
| 0x98209518 | 0x1f3a7518 | \SystemRoot\System32\Config\SAM                                      |
| 0x982d55c0 | 0x1e17c5c0 | \REGISTRY\MACHINE\BCD00000000                                        |
| 0xa620b008 | 0x1997b008 | ??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat      |
| 0xa65269c8 | 0x17ad09c8 | ??\C:\Users\sshd_server\ntuser.dat                                   |
| 0xa6539260 | 0x0f690260 | ??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat |
| 0xa7d2a9c8 | 0x01c469c8 | ??\C:\System Volume Information\Syscache.hve                         |

```
(root@kali)-[/home/kali/volatility]
#
```

Legend: (S) = Stable (V) = Volatile

```
Registry: \REGISTRY\MACHINE\HARDWARE
Key name: HARDWARE (S)
Last updated: 2019-01-06 18:02:40 UTC+0000
```

```
Subkeys:
(S) ACPI
(S) DESCRIPTION
(S) DEVICEMAP
```

```
Values:
```

```
(root@kali)-[/home/kali/volatility]
#
```

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\HARDWARE

Key name: 0 (S)

Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:

Values:

REG\_BINARY Component Information : (S)  
0x00000000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
REG\_SZ Identifier : (S) x86 Family 6 Model 158 Stepping 9  
REG\_FULL\_RESOURCE\_DESCRIPTOR Configuration Data : (S) ♦♦♦♦♦♦♦♦  
REG\_SZ ProcessorNameString : (S) Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz  
REG\_SZ VendorIdentifier : (S) GenuineIntel  
REG\_DWORD FeatureSet : (S) 2697805823  
REG\_DWORD ~MHz : (S) 3600  
REG\_DWORD Update Status : (S) 2  
REG\_BINARY Update Signature : (S)  
0x00000000 00 00 00 00 00 00 00 00 .....  
REG\_BINARY Previous Update Signature : (S)  
0x00000000 00 00 00 00 00 00 00 00 .....  
REG\_DWORD Platform ID : (S) 2

└─(root@kali)-[/home/kali/volatility]

Key name: System (S)

Last updated: 2019-01-06 18:02:44 UTC+0000

Subkeys:

(S) CentralProcessor  
(S) FloatingPointProcessor  
(S) MultifunctionAdapter

Values:

REG\_BINARY Component Information : (S)  
0x00000000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
REG\_SZ Identifier : (S) AT/AT COMPATIBLE  
REG\_FULL\_RESOURCE\_DESCRIPTOR Configuration Data : (S) ♦♦♦♦♦♦♦♦ ♦♦?♦  
REG\_SZ SystemBiosDate : (S) 06/23/99  
REG\_MULTI\_SZ SystemBiosVersion : (S) ['VBOS - 1', '', '']  
REG\_DWORD BootArchitecture : (S) 3  
REG\_DWORD PreferredProfile : (S) 0  
REG\_DWORD Capabilities : (S) 1345  
REG\_MULTI\_SZ VideoBiosVersion : (S) ['Oracle VM VirtualBox Version 6.0.0 VGA BIOS', 'Oracle VM VirtualBox Versi  
on 6.0.0 VGA BIOS', 'Oracle VM VirtualBox Version 6.0.0', 'Oracle VM VirtualBox Version 6.0.0', '', '', '', '',  
'', '', '', '', '', '', '', '', '', '']

└─(root@kali)-[/home/kali/volatility]

Legend: (S) = Stable (V) = Volatile

---

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: CMI-CreateHive{F10156BE-0E87-4EFB-969E-5DA29D131144} (S)

Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:

- (S) ControlSet001
- (S) ControlSet002
- (S) MountedDevices
- (S) RNG
- (S) Select
- (S) Setup
- (S) WPA
- (V) CurrentControlSet

Values:

```
(root@kali)-[/home/kali/volatility]
#
```

Legend: (S) = Stable (V) = Volatile

---

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: Enum (S)

Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:

- (S) ACPI
- (S) DISPLAY
- (S) HTREE
- (S) IDE
- (S) LPTENUM
- (S) PCI
- (S) PCIIDE
- (S) Root
- (S) SCSI
- (S) STORAGE
- (S) SW
- (S) USBSTOR
- (S) VMBUS
- (S) WpdBusEnumRoot

Legend: (S) = Stable (V) = Volatile

---

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: PCI (S)

Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:

- (S) VEN\_1000&DEV\_0054&SUBSYS\_1F091028&REV\_01
- (S) VEN\_1002&DEV\_515E&SUBSYS\_01E61028&REV\_02
- (S) VEN\_1022&DEV\_2000&SUBSYS\_20001022&REV\_40
- (S) VEN\_106B&DEV\_003F&SUBSYS\_00000000&REV\_00
- (S) VEN\_14E4&DEV\_1659&SUBSYS\_01E61028&REV\_11
- (S) VEN\_15AD&DEV\_0790&SUBSYS\_079015AD&REV\_02
- (S) VEN\_15AD&DEV\_07A0&SUBSYS\_07A015AD&REV\_01
- (S) VEN\_8086&DEV\_032C&SUBSYS\_00000000&REV\_09
- (S) VEN\_8086&DEV\_100E&SUBSYS\_001E8086&REV\_02
- (S) VEN\_8086&DEV\_1237&SUBSYS\_00000000&REV\_02
- (S) VEN\_8086&DEV\_265C&SUBSYS\_00000000&REV\_00
- (S) VEN\_8086&DEV\_2778&SUBSYS\_01E61028&REV\_00
- (S) VEN\_8086&DEV\_2779&SUBSYS\_01E61028&REV\_00
- (S) VEN\_8086&DEV\_27C8&SUBSYS\_01E61028&REV\_01
- (S) VEN\_8086&DEV\_27C9&SUBSYS\_01E61028&REV\_01
- (S) VEN\_8086&DEV\_27CA&SUBSYS\_01E61028&REV\_01
- (S) VEN\_8086&DEV\_27CC&SUBSYS\_01E61028&REV\_01

Legend: (S) = Stable (V) = Volatile

---

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: CurrentControlSet (V)

Last updated: 2019-01-06 18:02:40 UTC+0000

Subkeys:

Values:

REG\_LINK SymbolicLinkValue : (V) \Registry\Machine\System\ControlSet001

**(root@kali)-[/home/kali/volatility]**  
#

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: ComputerName (S)

Last updated: 2015-09-21 09:48:51 UTC+0000

Subkeys:

Values:

REG\_SZ : (S) mnmsrvc

REG\_SZ ComputerName : (S) IE8WIN7

(root@kali)-[/home/kali/volatility]  
#

(root@kali)-[/home/kali/volatility]  
# python2 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem --profile=Win7SP1x86\_23418 netscan | grep TCPv4

Volatility Foundation Volatility Framework 2.6.1

|            |       |                  |           |           |     |        |
|------------|-------|------------------|-----------|-----------|-----|--------|
| 0x21478bb0 | TCPv4 | 0.0.0.0:49155    | 0.0.0.0:0 | LISTENING | 432 | servic |
| es.exe     |       |                  |           |           |     |        |
| 0x269ff708 | TCPv4 | 0.0.0.0:49156    | 0.0.0.0:0 | LISTENING | 440 | lsass. |
| exe        |       |                  |           |           |     |        |
| 0x3e446008 | TCPv4 | 0.0.0.0:49154    | 0.0.0.0:0 | LISTENING | 880 | svchos |
| t.exe      |       |                  |           |           |     |        |
| 0x3e446d98 | TCPv4 | 0.0.0.0:49154    | 0.0.0.0:0 | LISTENING | 880 | svchos |
| t.exe      |       |                  |           |           |     |        |
| 0x3e46e0c0 | TCPv4 | 192.168.56.8:139 | 0.0.0.0:0 | LISTENING | 4   | System |
|            |       |                  |           |           |     |        |
| 0x3e671988 | TCPv4 | 0.0.0.0:445      | 0.0.0.0:0 | LISTENING | 4   | System |
|            |       |                  |           |           |     |        |
| 0x3e72e530 | TCPv4 | 0.0.0.0:135      | 0.0.0.0:0 | LISTENING | 672 | svchos |
| t.exe      |       |                  |           |           |     |        |
| 0x3e72ee48 | TCPv4 | 0.0.0.0:135      | 0.0.0.0:0 | LISTENING | 672 | svchos |
| t.exe      |       |                  |           |           |     |        |
| 0x3e736388 | TCPv4 | 0.0.0.0:49152    | 0.0.0.0:0 | LISTENING | 340 | winini |
| t.exe      |       |                  |           |           |     |        |
| 0x3e736d98 | TCPv4 | 0.0.0.0:49152    | 0.0.0.0:0 | LISTENING | 340 | winini |
| t.exe      |       |                  |           |           |     |        |
| 0x3e79d528 | TCPv4 | 0.0.0.0:49153    | 0.0.0.0:0 | LISTENING | 724 | svchos |

| Offset(V)  | Name   | PID | PPID | Thds | Hnds | Sess | Wow64 | Start                        | Exit |
|------------|--------|-----|------|------|------|------|-------|------------------------------|------|
| 0x8413a908 | System | 4   | 0    | 73   | 518  |      | 0     | 2019-01-06 18:02:44 UTC+0000 |      |

(root@kali)-[/home/kali/volatility]  
#

Usage: Volatility - A memory forensics analysis platform.

Options:

-h, --help list all available options and their default values.  
Default values may be set in the configuration file (/etc/volatilityrc)  
--conf-file=/root/.volatilityrc User based configuration file  
-d, --debug Debug volatility  
--plugins=PLUGINS Additional plugin directories to use (colon separated)  
--info Print information about all registered objects  
--cache-directory=/root/.cache/volatility Directory where cache files are stored  
--cache Use caching  
--tz=TZ Sets the (Olson) timezone for displaying timestamps using pytz (if installed) or tzset  
-f FILENAME, --filename=FILENAME Filename to use when opening an image  
--profile=WinXPSP2x86 Name of the profile to load (use --info to see a list of supported profiles)  
-l LOCATION, --location=LOCATION A URN location from which to load an address space  
-w, --write Enable write support  
--dtb=DTB DTB Address  
--shift=SHIFT Mac KASLR shift address

```
(root@kali)-[/home/kali/volatility]
# strings /home/kali/volatility3/volatility/memdumpWin7.mem | grep "192.168.56.5" > process_1.txt
```

```
(root@kali)-[/home/kali/volatility]
# ls -l process_1.txt
```

```
-rw-r--r-- 1 root root 13858 Sep 16 06:19 process_1.txt
```

```
(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# xxd process_1.txt | head -n 30
```

```
00000000: 2f73 6563 7265 745f 6669 6c65 2e64 6f63 /secret_file.doc
00000010: 7868 7474 7073 3a2f 2f77 7777 2e67 6f6f xhttps://www.goo
00000020: 676c 652e 636f 6d2f 7365 6172 6368 3f71 gle.com/search?q
00000030: 3d31 3932 2e31 3638 2e35 362e 3525 3246 =192.168.56.5%2F
00000040: 7365 6372 6574 5f66 696c 652e 646f 6378 secret_file.docx
00000050: 266f 713d 3139 322e 3136 382e 3536 2e35 &oq=192.168.56.5
00000060: 2532 4673 6563 7265 745f 6669 6c65 2e64 %2Fsecret_file.d
00000070: 6f63 7826 6171 733d 6368 726f 6d65 2e2e ocx&aqs=chrome..
00000080: 3639 6935 386a 3639 6935 3726 736f 7572 69i58j69i57&sour
00000090: 6365 6964 3d63 6872 6f6d 6526 6965 3d55 ceid=chrome&ie=U
000000a0: 5446 2d38 0a74 7470 3a2f 2f31 3932 2e31 TF-8.ttp://192.1
000000b0: 3638 2e35 362e 352f 7365 6372 6574 5f66 68.56.5/secret_f
000000c0: 696c 652e 646f 6378 0a68 7474 703a 2f2f ile.docx.http://
000000d0: 3139 322e 3136 382e 3536 2e35 2f73 6563 192.168.56.5/sec
000000e0: 7265 745f 6669 6c65 2e64 6f63 780a 7470 ret_file.docx.tp
000000f0: 3a2f 2f31 3932 2e31 3638 2e35 362e 352f ://192.168.56.5/
00000100: 0a09 6565 3462 6365 3530 2d31 6131 632d ..ee4bce50-1a1c-
00000110: 3439 6166 2d38 3165 372d 6438 6137 3138 49af-81e7-d8a718
00000120: 3037 3435 3836 3139 322e 3136 382e 3536 074586192.168.56
```

```

*****
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2???2
Cmd #37 @ 0x3200c4: 3?4?2???2
*****
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????

```

```

(root@kali)-[/home/kali/volatility]
#

```

| Offset(V)  | Name        | PID  | PPID | Thds | Hnds | Sess | Wow64 | Start                        | Exit |
|------------|-------------|------|------|------|------|------|-------|------------------------------|------|
| 0x842ddd28 | conhost.exe | 3988 | 348  | 2    | 52   | 1    | 0     | 2019-01-06 15:06:11 UTC+0000 |      |

```

(root@kali)-[/home/kali/volatility]
#

```

```

*****
ConsoleProcess: conhost.exe Pid: 2000
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0x54
-----
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
-----
CommandHistory: 0x3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
-----
Screen 0x3360a0 X:80 Y:300
Dump:
*****
ConsoleProcess: conhost.exe Pid: 3988
Console: 0xe181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4

```

```
OriginalTitle: %SystemRoot%\system32\cmd.exe
Title: C:\Windows\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0x5c
-----
CommandHistory: 0x1105d8 Application: ipconfig.exe Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x0
-----
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 at 0x10dc40: ipconfig
Cmd #1 at 0x107f90: cd Downloads
Cmd #2 at 0x1147a0: copy secret_file.docx F:
-----
Screen 0xf63b8 X:80 Y:300
Dump:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>ipconfig
```

```
Windows IP Configuration

Ethernet adapter Local Area Connection 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::1571:7b34:b2b1:bbe4%18
    IPv4 Address. . . . . : 192.168.56.8
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Tunnel adapter isatap.{7FD51246-C28D-46B6-AA98-EB2C150C388B}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter Teredo Tunneling Pseudo-Interface:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\IEUser>cd Downloads

C:\Users\IEUser\Downloads>copy secret_file.docx F:
```

```
1 file(s) copied.

C:\Users\IEUser\Downloads>

(root@kali)-[/home/kali/volatility]
#
```

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: DeviceClasses (S)

Last updated: 2019-01-06 15:03:08 UTC+0000

Subkeys:

- (S) {0e0b6031-5213-4934-818b-38d90ced39db}
- (S) {10497b1b-ba51-44e5-8318-a65c837b6661}
- (S) {2accfe60-c130-11d2-b082-00a0c91efb8b}
- (S) {2e34d650-5819-42ca-84ae-d30803bae505}
- (S) {32412632-86cb-44a2-9b5c-50d1417354f5}
- (S) {3375baf4-9e15-4b30-b765-67acb10d607b}
- (S) {34d14be3-dee4-41c8-9ae7-6b174977c192}
- (S) {35fa2e29-ea23-4236-96ae-3a6ebacba440}
- (S) {3abf6f2d-71c4-462a-8a92-1e6861e6af27}
- (S) {4116f60b-25b3-4662-b732-99a6111edc0b}
- (S) {4afa3d53-74a7-11d0-be5e-00a0c9062857}
- (S) {4d1e55b2-f16f-11cf-88cb-001111000030}
- (S) {4d36e978-e325-11ce-bfc1-08002be10318}
- (S) {53f56307-b6bf-11d0-94f2-00a0c91efb8b}
- (S) {53f56308-b6bf-11d0-94f2-00a0c91efb8b}
- (S) {53f5630d-b6bf-11d0-94f2-00a0c91efb8b}
- (S) {53f5630e-b6bf-11d0-94f2-00a0c91efb8b}
- (S) {53f56311-b6bf-11d0-94f2-00a0c91efb8b}

- (S) {57164f39-9115-4e78-ab55-382f3bd5422d}
- (S) {5b45201d-f2f2-4f3b-85bb-30ff1f953599}
- (S) {6ac27878-a6fa-4155-ba85-f98f491d4f33}
- (S) {72631e54-78a4-11d0-bcf7-00aa00b7b32a}
- (S) {866519b5-3f07-4c97-b7df-24c5d8a8ccb8}
- (S) {9527e630-d0ae-497b-adce-e80ab0175caf}
- (S) {97f76ef0-f883-11d0-af1f-0000f800845c}
- (S) {a9a0f4e7-5a45-4d96-b827-8a841e8c03e6}
- (S) {e849804e-c719-43d8-ac88-96b894c191e2}
- (S) {f33fdc04-d1ac-4e8e-9a30-19bbd4b108ae}

Values:

```
(root@kali)-[/home/kali/volatility]  
#
```

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: {53f56307-b6bf-11d0-94f2-00a0c91efb8b} (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

(S) ###IDE#DiskVBOX\_HARDDISK\_\_\_\_\_1.0\_\_\_\_\_#56106af1716061.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

(S) ###IDE#DiskVirtual\_HD\_\_\_\_\_1.1.0\_\_\_\_\_#5635dc70406060.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

(S) ###IDE#DiskVMware\_Virtual\_IDE\_Hard\_Drive\_\_\_\_\_00000001#562eba496060.0.0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

(S) ###SCSI#Disk&Ven\_Dell&Prod\_VIRTUAL\_DISK#6617b13437806000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

(S) ###USBSTOR#Disk&Ven\_General&Prod\_UDisk&Rev\_5.00#661bec0f48606\_60#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

Values:

(root@kali)-[/home/kali/volatility]

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: ###USBSTOR#Disk&Ven\_General&Prod\_UDisk&Rev\_5.00#661bec0f48606\_60#{53f56307-b6bf-11d0-94f2-00a0c91efb8b} (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

(S) #

(V) Control

Values:

REG\_SZ DeviceInstance : (S) USBSTOR\Disk&Ven\_General&Prod\_UDisk&Rev\_5.00\661bec0f48606\_60

(root@kali)-[/home/kali/volatility]

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: ###USBSTOR#Disk&Ven\_General&Prod\_UDisk&Rev\_5.00#661bec0f48606\_60#{53f56307-b6bf-11d0-94f2-00a0c91efb8b} (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

(S) #

(V) Control

Values:

REG\_SZ DeviceInstance : (S) USBSTOR\Disk&Ven\_General&Prod\_UDisk&Rev\_5.00\661bec0f48606\_60

(root@kali)-[/home/kali/volatility]

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: USBSTOR (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

(S) Disk&Ven\_General&Prod\_UDisk&Rev\_5.00

Values:

(root@kali)-[/home/kali/volatility]

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: Disk&Ven\_General&Prod\_UDisk&Rev\_5.00 (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

(S) 661bec0f48606\_60

Values:

(root@kali)-[/home/kali/volatility]  
#

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: 661bec0f48606\_60 (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

(S) Device Parameters

(S) LogConf

(S) Properties

(V) Control

Values:

REG\_SZ DeviceDesc : (S) @disk.inf,%disk\_devdesc%;Disk drive

REG\_DWORD Capabilities : (S) 0

REG\_MULTI\_SZ HardwareID : (S) ['USBSTOR\\DiskGeneral\_UDisk\_\_\_\_5.00', 'USBSTOR\\DiskGeneral\_UDisk\_\_\_\_', 'USBSTOR\\DiskGeneral\_', 'USBSTOR\\General\_UDisk\_\_\_\_5', 'General\_UDisk\_\_\_\_5', 'USBSTOR\\GenDisk', 'GenDisk', '', '']

REG\_MULTI\_SZ CompatibleIDs : (S) ['USBSTOR\\Disk', 'USBSTOR\\RAW', '', '']

REG\_SZ ContainerID : (S) {421e7e37-11dd-11e9-b3cf-806e6f6e6963}

REG\_DWORD ConfigFlags : (S) 0

REG\_SZ ClassGUID : (S) {4d36e967-e325-11ce-bfc1-08002be10318}

REG\_SZ Class : (S) DiskDrive

REG\_SZ Mfg : (S) @disk.inf,%genmanufacturer%;(Standard disk drives)

REG\_SZ Service : (S) disk

REG\_SZ FriendlyName : (S) General UDisk USB Device

Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM

Key name: MountedDevices (S)

Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

Values:

```
REG_BINARY    \DosDevices\C: : (S)
0x00000000    cd d7 ee 50 00 00 10 00 00 00 00 00 ... P.....
REG_BINARY    \??\Volume{a5b8a980-608c-11e5-a266-806e6f6e6963} : (S)
0x00000000    cd d7 ee 50 00 00 10 00 00 00 00 00 ... P.....
REG_BINARY    \DosDevices\D: : (S)

REG_BINARY    \DosDevices\E: : (S)
0x00000000    5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00 \.??.\.I.D.E.#.
0x00000010    43 00 64 00 52 00 6f 00 6d 00 56 00 42 00 4f 00 C.d.R.o.m.V.B.O.
0x00000020    58 00 5f 00 43 00 44 00 2d 00 52 00 4f 00 4d 00 X._.C.D.-.R.O.M.
0x00000030    5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 _._._._._._._._
0x00000040    5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 _._._._._._._._
0x00000050    5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 _._._._._._._._
0x00000060    5f 00 5f 00 5f 00 5f 00 5f 00 31 00 2e 00 30 00 _._._._._.1 ... 0.
0x00000070    5f 00 5f 00 5f 00 5f 00 5f 00 23 00 35 00 26 00 _._._._._.#.5.8.
0x00000080    33 00 39 00 34 00 63 00 30 00 61 00 64 00 33 00 3.9.4.c.0.a.d.3.
```

Legend: (S) = Stable (V) = Volatile

Registry: \??\C:\Users\IEUser\ntuser.dat

Key name: F (S)

Last updated: 2019-01-06 15:03:06 UTC+0000

Subkeys:

Values:

```
(root@kali)-[/home/kali/volatility]
#
```

```
*****
Process: 2524 explorer.exe
Cache type "URL " at 0x3156080
Record length: 0x200
Location: https://www.google.com/chrome/static/images/favicons/favicon.ico
Last modified: 2018-04-26 05:30:00 UTC+0000
Last accessed: 2018-09-17 14:54:34 UTC+0000
File Offset: 0x200, Data Offset: 0xac, Data Length: 0xbc
File: favicon[2].ico
Data: HTTP/1.1 200 OK
Content-Type: image/x-icon
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Alt-Svc: quic=":443"; ma=2592000; v="44,43,39,35"
Transfer-Encoding: chunked
```

~U:ieuser

```
*****
Process: 2524 explorer.exe
Cache type "URL " at 0x3156480
Record length: 0x180
Location: https://www.jam-software.de/css/reset_v4.css
Last modified: 2015-09-11 10:49:57 UTC+0000
```

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem --profile=Win7SP1x86_23418 iehistory | g
rep 2019
Volatility Foundation Volatility Framework 2.6.1
```

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem --profile=Win7SP1x86_23418 shellbags | g
rep "Last updated: 2019-01-06" | sort
Volatility Foundation Volatility Framework 2.6.1
Last updated: 2019-01-06 14:32:21 UTC+0000
Last updated: 2019-01-06 14:32:53 UTC+0000
Last updated: 2019-01-06 14:55:45 UTC+0000
Last updated: 2019-01-06 15:04:40 UTC+0000
Last updated: 2019-01-06 15:04:40 UTC+0000
Last updated: 2019-01-06 15:04:40 UTC+0000
Last updated: 2019-01-06 15:07:03 UTC+0000
```

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem --profile=Win7SP1x86_23418 shellbags --o
utput=html --output-file=shellbags.html
```

```
Volatility Foundation Volatility Framework 2.6.1
** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
** Failed to import volatility.plugins.ssd (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Scanning for registries...
Gathering shellbag items and building path tree...
Outputting to: shellbags.html
```

```
(root@kali)-[/home/kali/volatility]
# ls shellbags.html
```

shellbags.html

```
(root@kali)-[/home/kali/volatility]
```

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "
ControlSet001\Enum\USBSTOR"

Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssd (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Legend: (S) = Stable (V) = Volatile

Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
(S) Disk&Ven_General&Prod_UDisk&Rev_5.00

Values:

(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem --profile=Win7SP1x86_23418 printkey -K "
Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F"

Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssd (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Legend: (S) = Stable (V) = Volatile

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: F (S)
Last updated: 2019-01-06 15:03:06 UTC+0000

Subkeys:

Values:

(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem --profile=Win7SP1x86_23418 pslist -p 398
8

Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssd (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Offset(V) Name PID PPID Thds Hnds Sess Wow64 Start Exit

0x842ddd28 conhost.exe 3988 348 2 52 1 0 2019-01-06 15:06:11 UTC+0000

(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# python2 vol.py -f /home/kali/volatility3/volatility/memdumpWin7.mem --profile=Win7SP1x86_23418 hashdump

Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.ssd (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:fc525c9683e8fe067095ba2ddc971889:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
IEUser:1000:aad3b435b51404eeaad3b435b51404ee:fc525c9683e8fe067095ba2ddc971889:::
sshd_1001:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
sshd_server:1002:aad3b435b51404eeaad3b435b51404ee:8d0a16cfc061c3359db455d00ec27035:::

(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# echo fc525c9683e8fe067095ba2ddc971889 > ntlmhash.txt

(root@kali)-[/home/kali/volatility]
# ls -l

total 1292
-rw-rw-r-- 1 kali kali 778 Sep 14 11:38 AUTHORS.txt
-rw-rw-r-- 1 kali kali 23831 Sep 14 11:38 CHANGELOG.txt
drwxrwxr-x 4 kali kali 4096 Sep 14 11:38 contrib
-rw-rw-r-- 1 kali kali 3928 Sep 14 11:38 CREDITS.txt
-rw-r--r-- 1 root root 577307 Sep 16 06:26 index.html
-rw-r--r-- 1 root root 577306 Sep 16 06:27 index.html.1
-rw-rw-r-- 1 kali kali 698 Sep 14 11:38 LEGAL.txt
-rw-rw-r-- 1 kali kali 15127 Sep 14 11:38 LICENSE.txt
-rw-rw-r-- 1 kali kali 178 Sep 14 11:38 Makefile
-rw-rw-r-- 1 kali kali 348 Sep 14 11:38 MANIFEST.in
-rw-r--r-- 1 root root 33 Sep 16 07:42 ntlmhash.txt
-rw-rw-r-- 1 kali kali 254 Sep 14 11:38 PKG-INFO
-rw-r--r-- 1 root root 13858 Sep 16 06:19 process_1.txt
drwxrwxr-x 2 kali kali 4096 Sep 14 11:38 pyinstaller
-rw-rw-r-- 1 kali kali 1007 Sep 14 11:38 pyinstaller.spec
-rw-rw-r-- 1 kali kali 32164 Sep 14 11:38 README.txt
```

```
(root@kali)-[/home/kali/volatility]
# sudo find / -type f -name password.lst

find: '/run/user/1000/doc': Permission denied
find: '/run/user/1000/gvfs': Permission denied
/usr/share/john/password.lst
/usr/share/metasploit-framework/data/wordlists/password.lst
```

```
(root@kali)-[/home/kali/volatility]
# grep -i pass /usr/share/john/password.lst

#!comment: This list is based on passwords most commonly seen on a set of Unix
#!comment: (that is, more common passwords are listed first). It has been
#!comment: revised to also include common website passwords from public lists
#!comment: of "top N passwords" from major community website compromises that
password
password1
passion
Password
passw0rd
wordpass
password2
PASSWORD
newpass
passwd
nopass
pass

(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# echo Passw0rd! >> /usr/share/john/password.lst

(root@kali)-[/home/kali/volatility]
# grep Pass /usr/share/john/password.lst

Password
Passw0rd!

(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# john --format=nt ntlmhash.txt

Created directory: /root/.john
Using default input encoding: UTF-8
Loaded 1 password hash (NT [MD4 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=4
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
Passw0rd! (?)
1g 0:00:00:00 DONE 2/3 (2026-09-16 07:47) 100.0g/s 364800p/s 364800c/s 364800C/s !@#%$..sammy
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.

(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# git clone https://github.com/cube0x8/chrome_ragamuffin.git
Cloning into 'chrome_ragamuffin' ...
remote: Enumerating objects: 108, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 108 (delta 0), reused 0 (delta 0), pack-reused 105 (from 1)
Receiving objects: 100% (108/108), 220.24 KiB | 562.00 KiB/s, done.
Resolving deltas: 100% (51/51), done.
```

```
(root@kali)-[/home/kali/volatility]
# cd /home/kali/volatility
python2 vol.py --plugins=/home/kali/volatility/chrome_ragamuffin --info | grep -i -E "chrome|history"
Volatility Foundation Volatility Framework 2.6.1
chrome_ragamuffin - Recover some useful artifact from Chrome process memory
cmdscan - Extract command history by scanning for _COMMAND_HISTORY
consoles - Extract command history by scanning for _CONSOLE_INFORMATION
iehistory - Reconstruct Internet Explorer cache / history
linux_bash - Recover bash history from bash process memory
mac_bash - Recover bash history from bash process memory

(root@kali)-[/home/kali/volatility]
#
```

```
(root@kali)-[/home/kali/volatility]
# grep -n "class\|def calculate\|plugin" /home/kali/volatility/chrome_ragamuffin/chrome_ragamuffin.py | head -
30
27:import volatility.plugins.common as common
35:class DocumentFlagScanner(scan.ScannerCheck):
49:class NavigationControllerPendingTransient(scan.ScannerCheck):
67:class NavigationControllerEntriesCheck():
87:class DocumentScanner(scan.BaseScanner):
90:class NavigationControllerScanner(scan.BaseScanner):
93:class WhatsappScanner():
117:     if node.class_name == "chat-body":
142:class DOMScanner():
173:class _node(obj.CType):
198:class _element_mixin():
227:     def class_name(self):
228:         return libchrome.get_chrome_string(self, self.Element.m_elementData.class_names_.key_string_.v())
253:class _element_data(obj.CType):
258:class _element(obj.CType):
291:     def class_name(self):
292:         return libchrome.get_chrome_string(self, self.m_elementData.class_names_.key_string_.v())
318:class _html_element_form(_element_mixin, obj.CType):
328:class _html_iframe_element(_element_mixin, obj.CType):
344:class _html_anchor_element(_element_mixin, obj.CType):
349:class _html_image_element(_element_mixin, obj.CType):
```

```
(root@kali)-[/home/kali/volatility]
# echo -e "Index\tURL\t\t\tTitle\n-----\t\t\t\t\t" 66 strings /home/kali/volatility3/volatility/memdumpPW
in7.mem | grep -E "http://|https://" | grep -i "secret_file.docx" | awk '{print NR "\t" $1 "\tChrome Browser"}'
| head -n 10

Index    URL                                           Title
-----
1        /secret_file.docxhttps://www.google.com/search?q=192.168.56.5%2Fsecret_file.docx&oq=192.168.56.5%2Fsecre
t_file.docx&aqs=chrome..69i58j69i57&sourceid=chrome&ie=UTF-8 Chrome Browser
2        http://192.168.56.5/secret_file.docx Chrome Browser
3        86a18030-7004-41a1-bd3b-59814af00f23192.168.56.5/secret_file.docxhttp://192.168.56.5/secret_file.docxi,C
hrome Browser
4        /secret_file.docxhttps://www.google.com/search?q=192.168.56.5%2Fsecret_file.docx&oq=192.168.56.5%2Fsecre
t_file.docx&aqs=chrome..69i58j69i57&sourceid=chrome&ie=UTF-8 Chrome Browser
5        9bfdbe39-9f2d-4d9b-8a50-1c4844d4b5f7192.168.56.5/secret_file.docxhttps://www.google.com/search?q=192.168
.56.5%2Fsecret_file.docx&oq=192.168.56.5%2Fsecret_file.docx&aqs=chrome..69i58j69i57&sourceid=chrome&ie=UTF-8k C
hrome Browser
6        59d6c3cc-1751-4a78-9ecc-5708f93b956d192.168.56.5/secret_file.docxhttp://192.168.56.5/secret_file.docx C
hrome Browser
7        2890f4e1-0636-4d70-8975-d96f647aa899192.128.56.5/secret_file.docxhttps://www.google.com/search?q=192.128
.56.5%2Fsecret_file.docx&oq=192.128.56.5%2Fsecret_file.docx&aqs=chrome..69i58j69i57&sourceid=chrome&ie=UTF-8k C
hrome Browser
8        9354e24a-7bd6-4678-b729-076994f3a366192.128.56.5/secret_file.docxhttp://192.128.56.5/secret_file.docx C
```